



Comparative evaluation of micropropagated and conventionally propagated thornless blackberry (*Rubus* spp.) cultivars: Agronomic performance, fruit biochemical quality, and functional food potential

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ABSTRACT

Background: Blackberry (*Rubus* spp.) is a valuable functional food crop rich in phenolics, anthocyanins, flavonoids, vitamin C, and other bioactive compounds. Although micropropagation enables rapid production of uniform planting material, its effects on multi-year field performance and fruit biochemical quality remain insufficiently documented under specific agroecological conditions.

Objective: This study compared micropropagated (MP) and conventionally propagated (CP) plants of four thornless blackberry cultivars ('Doyle', 'Loch Ness', 'Natchez', and 'Freedom') under Armenian conditions, focusing on field performance, disease incidence, and fruit biochemical quality relevant to functional food applications.

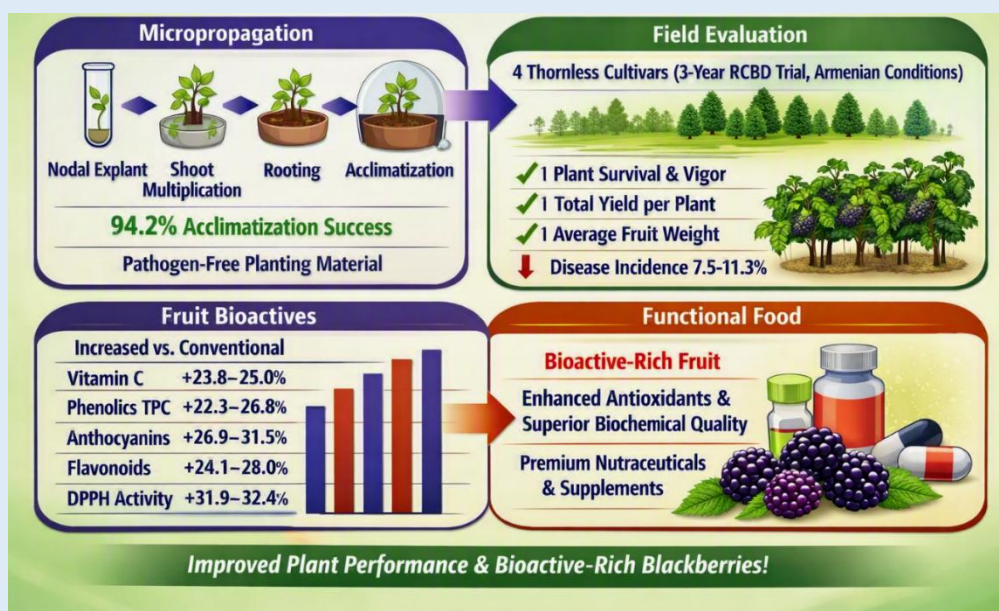
Methods: MP plants were produced through nodal explant culture, shoot multiplication, IBA-mediated rooting, and greenhouse acclimatization. Field trials were established in 2021 using a randomized complete block design. During 2022–2023, plant survival, growth, flowering, yield, fruit weight, and disease incidence were evaluated. Fruit quality was assessed for vitamin C, total phenolic content (TPC), anthocyanins, flavonoids, and DPPH antioxidant capacity.

Results: MP plants generally showed higher survival, vegetative growth, flowering, yield, fruit weight, and lower disease incidence than CP plants. In 2023, MP ‘Loch Ness’ produced 17.4 stems plant⁻¹, while disease incidence was 7.5–11.3% in MP versus 18.1–26.5% in CP plants. MP fruits contained higher levels of bioactive compounds, with the highest values in ‘Natchez’: vitamin C, 31.4 mg/100 g FW; TPC, 412.7 mg GAE/100 g FW; anthocyanins, 159.2 mg C3G/100 g FW; and flavonoids, 62.1 mg CE/100 g FW. DPPH antioxidant capacity was approximately 32% higher in MP fruits across all cultivars.

Conclusion: Micropropagation improved field performance, phytosanitary status, and fruit biochemical quality of the evaluated thornless blackberry cultivars. It therefore offers a promising strategy for standardized production of high-quality blackberry planting material and fruit with functional food potential.

Novelty of the study: This study provides a multi-season, cultivar-specific comparison of micropropagated and conventionally propagated thornless blackberry plants under identical open-field conditions in Armenia. By combining agronomic performance, disease incidence, and fruit biochemical traits, it evaluates how propagation method influences both field productivity and the accumulation of selected bioactive compounds.

Keywords: Blackberry; *Rubus* spp.; Micropropagation; Bioactive compounds; Antioxidant activity; Functional food



Graphical abstract: Systematic comparison of field performance, productivity, disease incidence, and fruit biochemical quality between MP and CP thornless blackberry cultivars.

INTRODUCTION

Blackberry (*Rubus* spp.) is an economically important berry crop rich in diverse bioactive compounds, including anthocyanins, ellagitannins, ellagic acid, flavonoids, phenolic acids, vitamin C, and dietary fiber [1,2]. Due to its nutritional and bioactive composition, blackberry may also be a promising raw material for functional food development [3,4]. Recent research has highlighted the nutritional and technological potential of blackberry-derived materials, including their use in value-added food products and seed-residue valorization [5]. These findings underscore the importance of factors affecting blackberry fruit composition and quality.

Blackberry phytochemical composition varies with genotype, environment, cultivation practices, fruit maturity, and plant status [1,2]. Thornless cultivars are commercially important because of their productivity, fruit quality, harvesting efficiency, and reduced labor requirements [6]. ‘Loch Ness’, ‘Natchez’, ‘Freedom’, and ‘Doyle’ are widely used thornless cultivars with distinct agronomic and fruit-quality traits.

Conventional propagation by stem cuttings and root suckers remains widely used but depends on propagation material quality, rooting capacity, environmental conditions, and mother-plant health [7]. Micropropagation provides a controlled alternative for rapid multiplication and production of healthy, uniform planting material, although cultivar-specific responses and genetic fidelity require consideration [8–11]. Clonal micropropagation may also affect subsequent plant performance and biochemical characteristics, as reported in other clonally propagated plant species [12].

Despite extensive research on blackberry fruit composition and micropropagation, comparative multi-season data on the field performance, productivity, disease incidence, and fruit biochemical quality of MP plants remain limited, particularly under Armenian agroecological conditions. Evaluating whether advantages of *in vitro* propagation persist after field establishment is therefore important for practical application.

Accordingly, this study compared MP and CP plants of four thornless blackberry cultivars under open-field conditions, assessing establishment, growth, flowering, yield, disease incidence, and fruit biochemical characteristics. We hypothesized that MP plants would show superior agronomic performance, lower disease incidence, and higher levels of selected bioactive compounds.

MATERIALS AND METHODS

Plant material and study site: The study was conducted at the Agrobiotechnology Research Center of the Armenian National Agrarian University (ANAU), Armenia, using four thornless blackberry (*Rubus* spp.) cultivars: ‘Doyle’, ‘Loch Ness’, ‘Natchez’, and ‘Freedom’. Micropropagation, including shoot initiation, multiplication, rooting, and acclimatization, was performed in 2021. Following acclimatization, uniform MP plants were transferred to the open field and compared with CP plants. Field trials were conducted in Gegakert village, Armavir Marz, during 2021–2023, with establishment, growth, flowering, yield, disease incidence, and fruit biochemical characteristics evaluated during 2022–2023. All plants received identical irrigation, fertilization, pruning, weed-control, and pest-management practices.

Explant sterilization: Nodal segments with a single axillary bud were collected from healthy mother plants and subjected to sterilization treatments using 70% ethanol for 30 s, 2% calcium hypochlorite [$\text{Ca}(\text{OCl})_2$] for 10–20 min, or a sequential treatment with 70% ethanol for 30 s followed by 2% calcium hypochlorite for 15–20 min.

***In vitro* shoot proliferation:** Sterile nodal explants (~2 cm) were cultured vertically on Murashige and Skoog (MS) medium [13] containing 3% (w/v) sucrose and 7 g L^{-1} agar (pH 5.7). Different concentrations of 6-benzylaminopurine (BAP), kinetin (Kin), and gibberellic acid (GA_3), ranging from 0.5 to 2.0 mg L^{-1} , and their

combinations were evaluated for shoot multiplication and elongation.

Each treatment included three independent replicates of 15 explants ($n = 45$). Cultures were maintained at 25 ± 2 °C under a 16 h photoperiod with cool-white fluorescent light ($40\text{--}50 \mu\text{mol m}^{-2} \text{s}^{-1}$). After 30 days, shoot number per explant and mean shoot length were recorded.

Root formation: Well-developed shoots (2–3 cm) were transferred to half-strength MS ($\frac{1}{2}$ MS) medium supplemented with IBA ($0.5\text{--}2.0 \text{ mg L}^{-1}$), with hormone-free $\frac{1}{2}$ MS as the control. Each treatment included three independent replicates of 15 shoots ($n = 45$). After 30 days, rooting percentage, number of roots per shoot, and mean root length were recorded. The optimal rooting response was obtained with 0.5 mg L^{-1} IBA.

Ex vitro acclimatization: Rooted plantlets were washed to remove residual agar and transplanted into pots containing perlite, forest soil, and biohumus (1:1:1, v/v/v). Acclimatization was conducted under greenhouse conditions, with relative humidity maintained at 90–95% during the first two weeks and gradually reduced over a five-week hardening period as light and ventilation were progressively increased [14–16]. Forty uniform and acclimatized MP plants per cultivar were selected for field evaluation.

Field trials: Forty plants per cultivar and propagation method were evaluated in a randomized complete block design (RCBD), with four blocks of 10 plants each. MP and CP plants were maintained under identical irrigation, fertilization, pruning, weed-control, and pest-management practices. Field performance was evaluated during 2022 and 2023, while 2021 was considered the establishment period and excluded from yield analysis. Plant survival, stem number, flower number, fruit yield, average fruit weight, and visible disease symptom incidence were recorded.

Disease incidence (%) = (Number of symptomatic plants / Total number of evaluated plants) $\times$ 100

Bioactive compound analysis: Vitamin C was determined by the 2,6-dichlorophenolindophenol (DCPIP) titrimetric method [17] and expressed as mg ascorbic acid per 100 g fresh weight (FW). TPC was determined using the Folin–Ciocalteu assay [18] and expressed as mg gallic acid equivalents (GAE) per 100 g FW. Total anthocyanins were determined by the pH differential method [19] and expressed as mg cyanidin-3-glucoside equivalents (C3G) per 100 g FW. Total flavonoids were determined using the aluminum chloride assay [20] and expressed as mg catechin equivalents (CE) per 100 g FW. Antioxidant capacity was assessed using the DPPH assay [21] and expressed as μmol Trolox equivalents (TE) per g FW.

Statistical analysis: *In vitro* experiments included three independent replicates, while field trials used an RCBD with four blocks and 10 plants per treatment per block. Biochemical analyses included four biological replicates per cultivar $\times$ propagation method combination, with each sample analyzed in triplicate.

Data are presented as mean $\pm$ standard error (SE). The effects of cultivar, propagation method, and their interaction were assessed using two-way ANOVA, followed by Tukey's HSD test for mean comparisons at $p < 0.05$.

RESULTS AND DISCUSSION

Establishment of micropropagated blackberry plants

and field adaptation: Nodal explants of the four thornless blackberry cultivars were successfully established *in vitro*. The most effective sterilization treatment (70% ethanol for 30 s followed by 2% calcium hypochlorite for 15–20 min) yielded >80% survival and 11.7–18.3% contamination. Prolonged sterilization reduced viability, while cultivar-dependent responses confirmed the influence of genotype on explant sensitivity. These results agree with previous reports on optimizing sterilization conditions in *Rubus* [8,10]. The

most effective combination of BAP, kinetin, and GA₃ (0.5, 0.5, and 1.0 mg L⁻¹, respectively) produced 7–9 shoots per explant, with mean shoot lengths of 9–12 cm, whereas hormone-free MS medium resulted in limited shoot development, supporting the importance of cytokinin balance in *Rubus* shoot proliferation [8,13]. Rooting was most effective on ½ MS medium supplemented with 0.5 mg L⁻¹ IBA, resulting in 80.0–86.7% rooting, 4–5 roots per shoot, and a mean root length of 9–11 cm, consistent with previous reports [8,10,13]. Rooted plantlets were successfully acclimatized in a mixture of perlite, forest soil, and biohumus (1:1:1, v/v/v). Gradual reduction of humidity, along with increased ventilation and light, facilitated the transition from *in vitro* to *ex vitro* conditions [14–16]. These results demonstrate the effectiveness of the established clonal micropropagation protocol in producing healthy and uniform plant material suitable for subsequent biochemical evaluation [12].

Overall, the protocol provided uniform plants for field evaluation and highlighted the importance of genotype-specific micropropagation protocols for blackberry production.

Field performance, vegetative growth, and yield characteristics of blackberry cultivars:

Field evaluation during 2022–2023 showed significant differences between CP and MP plants across the four thornless cultivars: ‘Doyle’, ‘Loch Ness’, ‘Natchez’, and ‘Freedom’ (Tables 1 and 2). Two-way ANOVA indicated significant effects of propagation method, cultivar, and their interaction on several agronomic traits ($p < 0.05$). Overall, MP plants showed better field establishment, vegetative growth, flowering, and yield than CP plants, with lower disease incidence. In 2022, all 40 MP plants per cultivar survived (100%), whereas 37–39 of 40 CP plants survived, corresponding to 92.5–97.5% survival (Table 1).

Table 1. Field performance of CP and MP thornless blackberry cultivars in 2022

Cultivar	Propagation type	Surviving plants	Disease incidence (%)	Stems plant ⁻¹	Flowers plant ⁻¹	Yield plant ⁻¹ (kg)	Fruit weight (g)
Doyle	Conventional	37	14.3 ± 0.38 ^b	3.7 ± 0.2 ^{cd}	215 ± 16 ^d	1.15 ± 0.11 ^d	6.65 ± 1.20 ^c
	Micropropagated	40	2.1 ± 0.31 ^e	5.1 ± 0.22 ^c	338 ± 23 ^{bc}	2.10 ± 0.22 ^{bc}	8.20 ± 1.55 ^b
Loch Ness	Conventional	37	18.7 ± 2.8 ^a	3.8 ± 0.9 ^{cd}	273 ± 21 ^{cd}	1.18 ± 0.12 ^d	5.65 ± 0.55 ^c
	Micropropagated	40	3.9 ± 0.51 ^e	7.4 ± 1.6 ^{ab}	449 ± 64 ^a	2.85 ± 0.31 ^{ab}	8.25 ± 1.25 ^b
Natchez	Conventional	38	14.6 ± 1.8 ^b	4.4 ± 1.3 ^c	254 ± 26 ^{cd}	1.72 ± 0.18 ^c	9.15 ± 1.20 ^{ab}
	Micropropagated	40	4.3 ± 1.1 ^{de}	8.5 ± 2.0 ^a	427 ± 53 ^{ab}	3.15 ± 0.32 ^a	9.85 ± 1.15 ^a
Freedom	Conventional	39	21.1 ± 2.0 ^a	2.4 ± 0.8 ^d	246 ± 27 ^{cd}	1.52 ± 0.16 ^{cd}	8.30 ± 0.90 ^b
	Micropropagated	40	4.8 ± 1.4 ^d	6.4 ± 1.9 ^b	484 ± 33 ^a	3.20 ± 0.28 ^a	9.35 ± 0.75 ^{ab}

Note: Values are mean ± SE for 2022. Surviving plants indicate the number of plants surviving per treatment (out of 40). Different lowercase letters indicate significant differences among cultivar × propagation method combinations, based on two-way ANOVA followed by Tukey’s HSD test at $p \leq 0.05$.

A similar pattern was observed in 2023, when all MP treatments had 40 surviving plants, compared with 35–37 surviving CP plants (Table 2). The higher establishment rate of MP plants indicates more consistent field establishment under the evaluated conditions. Micropropagation consistently enhanced vegetative growth. In 2022, MP plants produced 5.1–8.5 stems plant⁻¹ versus 2.4–4.4 in CP plants, with the largest difference in ‘Natchez’ (8.5 vs. 4.4 stems plant⁻¹) (Table 1). This advantage persisted in 2023, with 12.4–17.4 stems plant⁻¹ in MP versus 9.3–15.3 in CP plants; MP ‘Loch Ness’ recorded the highest value (17.4 stems plant⁻¹) (Table 2). Flowering was also greater in MP plants, reaching 338–484 flowers plant⁻¹ in 2022 and 416–544 in 2023, compared with 215–273 and 281–361

flowers plant⁻¹ in CP plants, respectively (Tables 1 and 2). This greater flowering intensity likely contributed to higher yields. Yield was consistently higher in MP plants, ranging from 2.10–3.20 kg plant⁻¹ in 2022 and 3.82–4.40 kg plant⁻¹ in 2023, compared with 1.15–1.72 and 2.50–3.09 kg plant⁻¹ in CP plants, respectively (Tables 1 and 2). The highest 2023 yields were recorded for MP ‘Loch Ness’ (4.40 kg plant⁻¹) and ‘Natchez’ (4.24 kg plant⁻¹). Fruit weight showed a similar trend, with MP fruits weighing 8.20–9.85 g in 2022 and 9.20–10.55 g in 2023, compared with 5.65–9.15 and 7.15–8.95 g in CP plants, respectively (Tables 1 and 2). ‘Natchez’ produced the heaviest fruits among the MP plants, while its fruit weight under CP was comparable to that of ‘Freedom’.

Table 2. Field performance of CP and MP thornless blackberry cultivars in 2023

Cultivar	Propagation type	Surviving plants	Disease incidence (%)	Stems plant ⁻¹	Flowers plant ⁻¹	Yield plant ⁻¹ (kg)	Fruit weight (g)
Doyle	Conventional	36	18.1 ± 1.9 ^b	10.4 ± 1.7 ^{cd}	281 ± 28 ^e	3.04 ± 0.22 ^c	7.65 ± 1.20 ^c
	Micropropagated	40	7.5 ± 1.1 ^d	15.3 ± 2.1 ^{ab}	432 ± 34 ^{bc}	3.82 ± 0.26 ^{ab}	9.20 ± 1.05 ^b
Loch Ness	Conventional	35	26.5 ± 2.8 ^a	15.3 ± 1.9 ^{ab}	361 ± 42 ^{de}	2.65 ± 0.31 ^d	7.15 ± 1.05 ^c
	Micropropagated	40	11.1 ± 1.7 ^c	17.4 ± 2.4 ^a	544 ± 43 ^a	4.40 ± 0.33 ^a	9.50 ± 1.75 ^b
Natchez	Conventional	37	24.5 ± 3.0 ^a	12.3 ± 2.2 ^{bc}	332 ± 31 ^{de}	3.09 ± 0.26 ^c	8.95 ± 0.85 ^b
	Micropropagated	40	11.3 ± 1.8 ^c	16.5 ± 1.9 ^a	511 ± 48 ^{ab}	4.24 ± 0.30 ^a	10.55 ± 1.50 ^a
Freedom	Conventional	37	19.4 ± 2.4 ^b	9.3 ± 1.8 ^d	320 ± 36 ^{de}	2.50 ± 0.26 ^d	8.95 ± 0.80 ^b
	Micropropagated	40	10.3 ± 1.2 ^c	12.4 ± 2.0 ^{bc}	416 ± 39 ^c	3.84 ± 0.24 ^{ab}	10.10 ± 0.75 ^{ab}

Note: Values are mean ± SE for 2023. Surviving plants indicate established plants from 40 transplanted plants per treatment. Different lowercase letters indicate significant differences among cultivar × propagation method combinations based on two-way ANOVA followed by Tukey’s HSD test at $p \leq 0.05$.

In addition to improved growth and productivity, MP plants showed a better phytosanitary status. Disease incidence was consistently lower in MP plants across all cultivars and years, ranging from 2.1–4.8% in 2022 and 7.5–11.3% in 2023, compared with 14.3–21.1% and 18.1–26.5% in CP plants, respectively (Tables 1 and 2). The lower disease incidence in MP plants may be related to

more uniform establishment and higher planting material quality, although the mechanisms underlying this difference were not directly evaluated in the present study.

Overall, the two-year evaluation demonstrated that MP thornless blackberry plants had better establishment, vegetative and reproductive performance, yield, and

phytosanitary status than CP plants under Armenian continental agroecological conditions. These findings support micropropagation as an efficient approach for establishing productive and uniform blackberry plantations under the evaluated conditions.

Bioactive compound accumulation and antioxidant capacity of blackberry fruits:

Biochemical evaluation showed that the propagation method significantly influenced bioactive compound accumulation and antioxidant capacity. MP fruits generally had higher levels of vitamin C, TPC, anthocyanins, flavonoids, and higher antioxidant capacity than CP fruits (Table 3), supporting their biochemical quality and potential for functional food applications [22–25]. Recent research on blackberry-derived materials further highlights the importance of their nutritional and bioactive composition for value-added food products [5]. Two-way ANOVA showed significant effects of cultivar, propagation method, and their interaction on most biochemical

parameters ($p < 0.05$), indicating cultivar-dependent responses. MP fruits had higher vitamin C levels across all cultivars, with the highest value in ‘Natchez’ (31.4 mg/100 g FW), followed by ‘Doyle’ (28.6), ‘Freedom’ (26.8), and ‘Loch Ness’ (24.1 mg/100 g FW), compared with 18.5–25.1 mg/100 g FW in CP fruits (Table 3). TPC followed a similar pattern, reaching 412.7 mg GAE/100 g FW in MP ‘Natchez’ and 295.0 mg GAE/100 g FW in CP ‘Loch Ness’. Phenolic and vitamin C profiles, as well as biochemical responses of regenerated plants, can be influenced by genotype, environmental conditions, and propagation conditions [26–28]. MP ‘Natchez’ fruits had the highest anthocyanin (159.2 mg C3G/100 g FW) and flavonoid (62.1 mg CE/100 g FW) contents, while CP fruits generally showed lower values (Table 3). This enhanced accumulation may be associated with the improved field establishment and vegetative performance observed in MP plants, although the physiological mechanisms underlying these differences were not directly evaluated.

Table 3. Bioactive compound composition and antioxidant capacity of blackberry fruits under different propagation methods

Cultivar	Propagation type	Vitamin C (mg/100 g FW)	TPC (mg GAE/100 g FW)	Anthocyanins (mg C3G/100 g FW)	Flavonoids (mg CE/100 g FW)	DPPH radical scavenging capacity (μ mol TE/g FW)
Doyle	Conventional	22.4 $\pm$ 1.1 ^c	310.5 $\pm$ 12.4 ^{cd}	112.3 $\pm$ 5.1 ^{de}	42.6 $\pm$ 1.8 ^{de}	14.2 $\pm$ 0.6 ^d
	Micropropagated	28.6 $\pm$ 1.3 ^{ab}	385.2 $\pm$ 15.1 ^{ab}	141.5 $\pm$ 6.3 ^{bc}	56.4 $\pm$ 2.1 ^b	18.8 $\pm$ 0.8 ^b
Loch Ness	Conventional	18.5 $\pm$ 0.9 ^d	295.0 $\pm$ 10.8 ^d	98.4 $\pm$ 4.2 ^e	38.2 $\pm$ 1.5 ^e	12.8 $\pm$ 0.5 ^d
	Micropropagated	24.1 $\pm$ 1.0 ^{bc}	360.8 $\pm$ 14.3 ^{bc}	128.0 $\pm$ 5.9 ^{cd}	49.8 $\pm$ 1.9 ^{bc}	16.9 $\pm$ 0.7 ^{bc}
Natchez	Conventional	25.1 $\pm$ 1.2 ^{bc}	340.2 $\pm$ 13.6 ^{bc}	125.6 $\pm$ 5.8 ^{cd}	48.5 $\pm$ 2.2 ^{cd}	15.8 $\pm$ 0.7 ^{cd}
	Micropropagated	31.4 $\pm$ 1.4 ^a	412.7 $\pm$ 16.8 ^a	159.2 $\pm$ 7.1 ^a	62.1 $\pm$ 2.5 ^a	20.9 $\pm$ 0.9 ^a
Freedom	Conventional	20.3 $\pm$ 0.8 ^{cd}	325.4 $\pm$ 11.9 ^{cd}	105.1 $\pm$ 4.7 ^e	41.0 $\pm$ 1.6 ^{de}	13.5 $\pm$ 0.5 ^d
	Micropropagated	26.8 $\pm$ 1.1 ^{ab}	394.1 $\pm$ 14.9 ^{ab}	136.4 $\pm$ 6.0 ^{bc}	53.7 $\pm$ 2.0 ^b	17.8 $\pm$ 0.7 ^{bc}

Note: Values are mean $\pm$ SE. Different lowercase letters within a column indicate significant differences among cultivar $\times$ propagation method combinations according to two-way ANOVA followed by Tukey’s HSD test at $p \leq 0.05$. FW, GAE; C3G; CE; TE.

Recent research has highlighted the nutritional and functional relevance of berry-derived bioactive compounds [29]. Reviews have summarized the occurrence and potential health relevance of phenolic compounds in fruits and other plant foods [30]. Studies of *Rubus* accessions have documented variation in anthocyanins, phenolics, flavonoids, and antioxidant activity [31], while metabolomic and biological-activity studies of blackberry cultivars further demonstrate cultivar-dependent biochemical profiles [32]. Berries are also recognized as nutritionally valuable food materials with potential for value-added processing [33]. Comparative studies of blackberry and raspberry genotypes further highlight variation in nutritional and antioxidant profiles [34]. Accordingly, the higher concentrations observed in MP fruits indicate an enhanced biochemical profile and may support their value as nutritionally valuable raw materials. However, these findings should not be interpreted as direct evidence of specific health or clinical effects.

Fruit antioxidant capacity, determined by the DPPH radical-scavenging assay, was consistently higher in MP than CP fruits (Table 3), with the highest activity in MP 'Natchez' (20.9 $\mu\text{mol TE/g FW}$) and the lowest in CP 'Loch Ness' (12.8 $\mu\text{mol TE/g FW}$). This higher antioxidant capacity corresponded with increased concentrations of phenolics, anthocyanins, flavonoids, and vitamin C, with DPPH activity approximately 32% higher across all four cultivars. Similar relationships between phenolic composition and antioxidant activity have been reported in blackberry and other *Rubus* species, with genotype strongly influencing antioxidant potential [32–34]. The higher biochemical values observed in MP fruits may be associated with better establishment, vegetative vigor, and growth uniformity, although the underlying mechanisms were not directly evaluated. Bioactive compound accumulation is also influenced by genotype and environmental conditions [27,28]. Overall,

micropropagation was associated with improved field performance, yield, fruit nutritional composition, and antioxidant potential, supporting its use for standardized production of high-quality blackberry fruits for functional food applications.

Alignment with the FFC 17-step functional food development framework: The FFC 17-Step Functional Food Development Framework provides a structured pathway for developing scientifically validated functional foods [4]. The present study addresses its early stages through standardized propagation, field evaluation, and biochemical characterization of four thornless blackberry (*Rubus* spp.) cultivars: 'Doyle', 'Loch Ness', 'Natchez', and 'Freedom'.

Rubus and other berries are rich in nutrients and bioactive compounds, with phytochemical profiles influenced by genotype and cultivation conditions [22–23]. Recent research has emphasized the importance of fruit quality and biochemical characterization in evaluating berry crops for functional food applications [35]. Accordingly, MP and CP plants were compared under identical field conditions.

Micropropagation enables standardized production under controlled *in vitro* conditions [9,10] and has been successfully applied to blackberry multiplication, rooting, acclimatization, and genetic uniformity [9–11]. In strawberries, *in vitro* propagation has likewise been associated with differences in field performance, yield, and fruit quality [35]. In the present study, MP plants showed improved field performance, productivity, and fruit biochemical quality compared with CP plants.

Biochemical assessment included vitamin C, phenolics, anthocyanins, flavonoids, and DPPH antioxidant capacity, which are commonly used indicators of the nutritional and antioxidant quality of berries [22,24]. MP fruits showed higher levels of these compounds and greater antioxidant capacity than CP

fruits, although these characteristics remain dependent on genotype and environmental conditions [22,27]. The study does not address later FFC stages, including individual compound identification, mechanisms of action, effective doses, bioavailability, or preclinical and clinical validation [4]. Similar considerations have been emphasized in recent research on plant-derived bioactive resources, where biochemical characterization and cultivation potential are considered important preliminary steps toward functional food applications [36]. Thus, the observed biochemical differences indicate enhanced fruit quality and functional food potential rather than demonstrated health efficacy.

Overall, micropropagation supports standardized blackberry production and quality characterization within the FFC framework [4, 9–11], providing a basis for further functional food applications.

Scientific innovations: The scientific novelty of this study lies in the direct two-season comparison of micropropagated and conventionally propagated thornless blackberry cultivars under identical open-field conditions in Armenia. By integrating field establishment, vegetative growth, flowering, yield, disease incidence, and fruit biochemical characteristics, the study provides cultivar-specific evidence that micropropagated plants can achieve comparable or improved field performance while maintaining desirable bioactive compound profiles. These findings support micropropagation as a potential strategy for producing uniform and productive blackberry planting material under Armenian field conditions.

Practical implications: Micropropagation is a practical tool for producing uniform and productive blackberry plants. Cultivar variation aids genotype selection, while higher bioactive levels in MP fruits indicate potential for quality-oriented berry production and functional food applications.

CONCLUSION

Under Armenian agroecological conditions, micropropagation improved the establishment, vegetative growth, flowering, yield, fruit quality, and phytosanitary status of thornless blackberry cultivars compared with conventional propagation. The optimized $\frac{1}{2}$ MS medium supplemented with 0.5 mg L^{-1} IBA provided effective rooting and vigorous plantlet development. Across two production seasons, micropropagated plants consistently showed higher survival, productivity, and lower disease incidence, while their fruits generally contained higher levels of bioactive compounds and antioxidant capacity. These results demonstrate that micropropagation is a promising approach for producing uniform, vigorous, and high-performing blackberry planting material under Armenian conditions.

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Abbreviations: BAP, 6-benzylaminopurine; CP, conventionally propagated; FW, fresh weight; GA_3 , gibberellic acid; GAE, gallic acid equivalents; IBA, indole-3-butyric acid; Kin, kinetin; MP, micropropagated; MS, Murashige and Skoog medium; RCBD, randomized complete block design; TPC, total phenolic content.

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